

lecture notes 22 (continued)

Really great MCMC reference! (long HW, implement your own MCMC by following these notes):

<https://arxiv.org/abs/1710.06068>

MCMC chain intuition building

We want our MCMC chains to look like **fuzzy caterpillars**, not **wiggly worms**. In other words, basically you want them to look like white noise! (Small autocorrelation length)

Signs of bad chains include lots of rejected proposals (streaks), or too small or too big jumps (wormy chains).

Some bad chain examples:

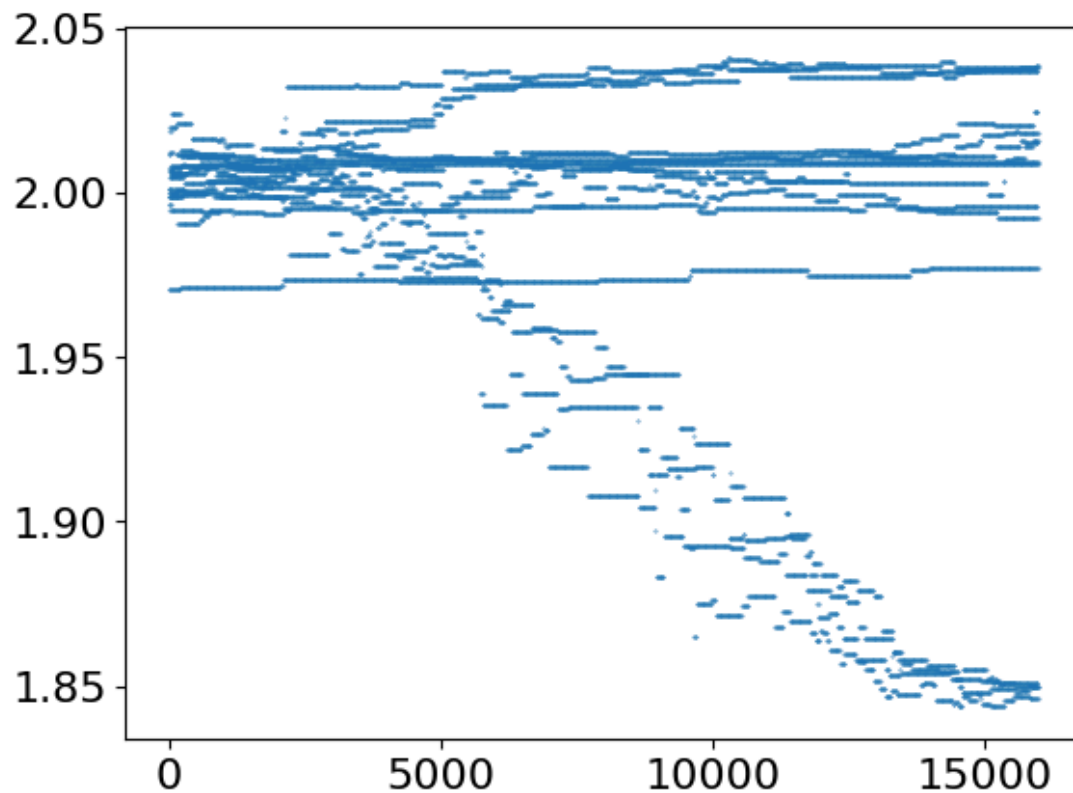


Figure 1: 46584740d2ee04390d0e5fc9bd167b70.png

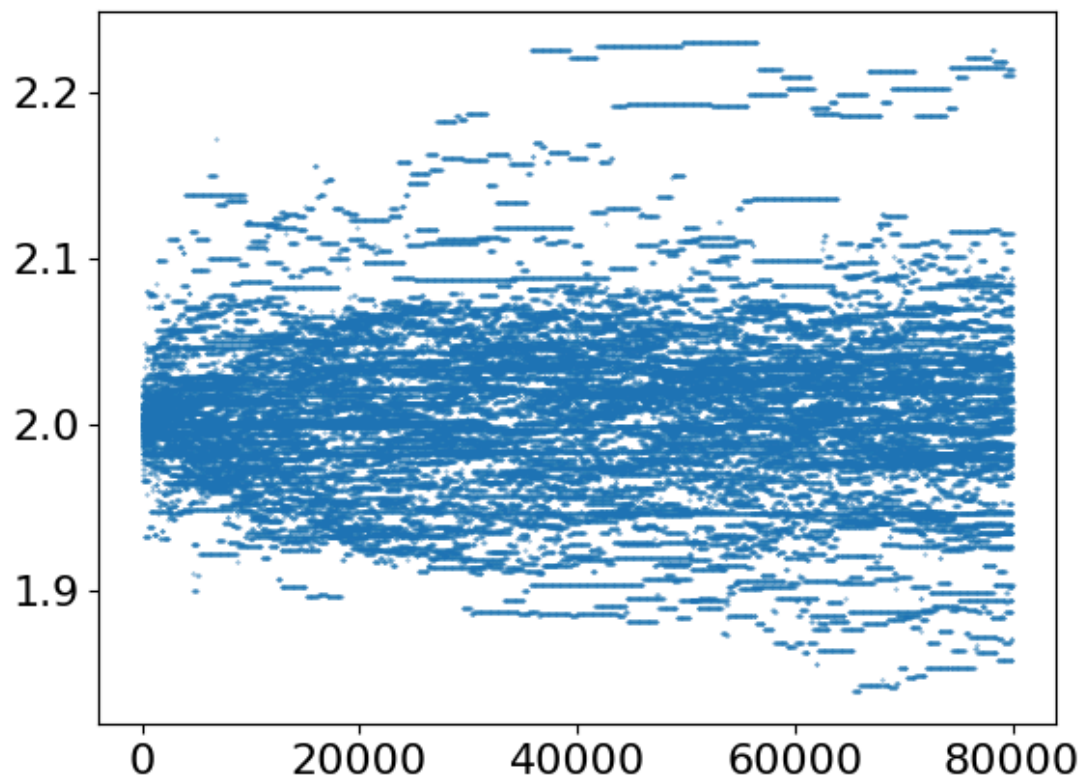


Figure 2: fe3f7c8b8a9f6b99281eed6e0927135.png

Some good chain examples:

